

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (CL) Examination

November 2013

CL302.01 Structural Analysis-II

Date: 20.11.2013, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

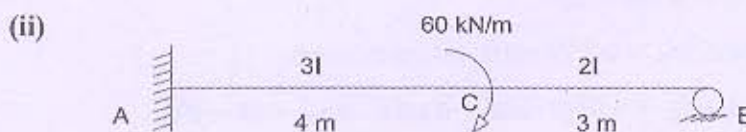
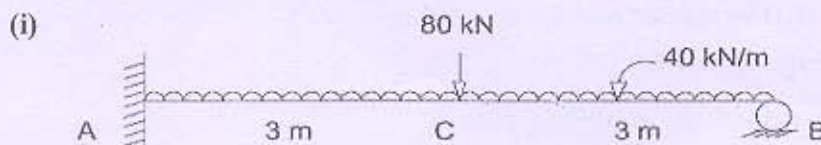
SECTION - I

Q - 1 A cable loaded with 10 kN/m is stretched between two supports in the same horizontal line [07]
200m apart. If the central dip is 15m, find the maximum and minimum pulls in the cables.

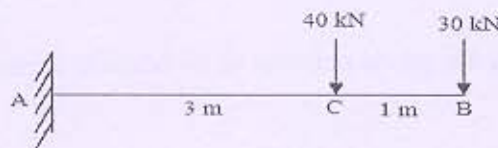
Q - 2 (a) Determine the strain energy of a cantilever beam of span 2 m having size 20 mm width x 60 [07]
mm depth. Take $E = 200$ GPa.

- (a) When 1000 N concentrated load is placed at free end.
- (b) When total 1000 N load is uniformly distributed over the entire span.

(b) Analyze the beam shown in Figures by consistent deformation method and draw S.F. [07]
and B.M. diagrams. **Attempt any One.**

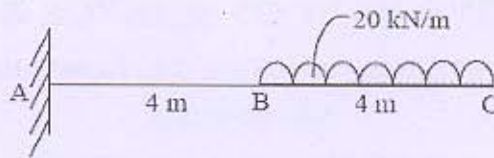


Q - 3 (a) Using energy principles, calculate θ_B and δ_B for a beam shown in Figure. Take $E = 2 \times 10^5$ [05]
 N/mm^2 , $I = 5 \times 10^8 \text{ mm}^4$.

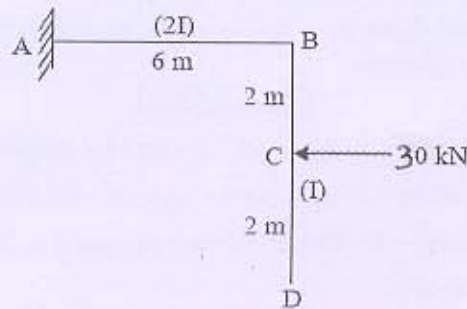


(b) Attempt any One. [09]

- (i) Determine deflection at the free end of a cantilever beam using unit load method. Take $E = 2 \times 10^5 \text{ N/mm}^2$, $I = 8 \times 10^8 \text{ mm}^4$.



- (ii) Determine horizontal deflection of point D for the frame shown in figure. Take $E = 2 \times 10^5$ N/mm², $I = 2 \times 10^8$ mm⁴.



SECTION - II

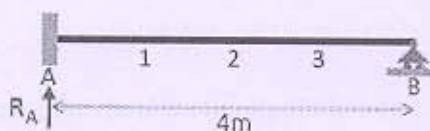
- Q - 4 A beam circular in plan is located uniformly with U.D.L. of 125 kN/m inclusive of self-weight. The radius of beam is 5m. The beam is symmetrically supported on five columns. Draw S.F., B.M. and T.M. (Use relevant equation from below) [07]

1. Shear Force at Support: $F_0 = wR\theta$
2. Bending Moment at Support: $M_0 = wR^2[1 - \theta \cot \theta]$
3. Shear Force at any Point: $F_\phi = wR[\theta - \phi]$
4. Bending Moment at any Point: $M_\phi = wR^2[\theta \sin \phi + \theta \cot \theta \cdot \cos \phi - 1]$
5. Twisting Moment at any Point: $T_\phi = wR^2[\theta \cos \phi - \theta \cot \theta \cdot \sin \phi - (\theta - \phi)]$
6. Position of Maximum Torque: $\sin \phi = \frac{1}{\theta} [\sin^2 \theta \pm \cos \theta \cdot (\theta^2 - \sin^2 \theta)^{\frac{1}{2}}]$

- Q - 5 (a) Define Flexibility & Stiffness. Write four characteristics of Stiffness Matrix. [04]

- (b) Attempt any One. [10]

- (i) Sketch ILD for R_A , R_B , M_A giving ordinates at 1m intervals for the propped cantilever shown in the Figure.



- (ii) Sketch ILD for M_B giving ordinates at 1m intervals for the two span continuous beam shown

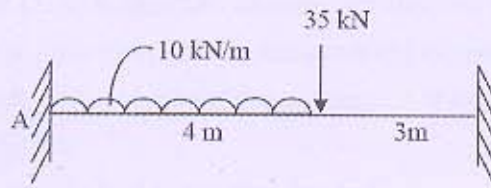
in the Figure. Assume EI is constant throughout.



Q - 6 (a) A propped cantilever beam of span 4m carries a point load of 75 kN at a distance 1m from fixed end. Find out support reactions and plot shear force and bending moment diagrams. Use **Flexibility or Stiffness method**. [05]

(b) Find out support reactions and draw shear force and bending moment diagrams. [09]
(Attempt any One)

(i)



(ii)

